

NETWORK SECURITY & SCANNING

ApexPlanet Cyber Security Internship

Task 2 Report

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ABSTRACT

This report documents the successful completion of **Task 2 – Network Security & Scanning** under the **ApexPlanet Cyber Security Internship** program. The primary objective of this task was to develop practical skills in **network reconnaissance, service enumeration, traffic analysis, firewall configuration, and vulnerability assessment** by working in a controlled cybersecurity lab environment.

A virtual lab was established using **Oracle VirtualBox**, where **Kali Linux** was configured as the attacker machine and **Metasploitable2** was deployed as the target machine. The task involved performing both **passive and active reconnaissance** to gather information about the target system, followed by systematic **network scanning and service identification** using **Nmap**. Various scanning techniques such as **host discovery, TCP SYN scanning, service version detection, and operating system detection** were used to identify open ports, exposed services, and the overall attack surface of the target machine.

To extend the analysis beyond port scanning, **Wireshark** was used to capture and inspect network packets, enabling observation of protocols such as **HTTP, FTP, DNS, and TCP**. This provided practical understanding of how network traffic flows between systems and how insecure or unencrypted communication can be analyzed. In addition, **firewall concepts** were explored through **iptables** to understand packet filtering, access control, and the role of basic defensive mechanisms in Linux-based systems.

The task also included **vulnerability assessment** using **OpenVAS (Greenbone Vulnerability Manager)**, which was configured to evaluate the target virtual machine for known weaknesses. The generated findings highlighted potential vulnerabilities and demonstrated how automated scanning tools assist security professionals in identifying, categorizing, and prioritizing risks based on severity levels.

Overall, this task provided valuable hands-on exposure to essential cybersecurity practices including **reconnaissance, network scanning, packet analysis, firewall management, and vulnerability assessment**. It strengthened practical knowledge of offensive and defensive security concepts while improving technical proficiency with industry-standard tools used in real-world security testing and network analysis.

1. INTRODUCTION

In today's digital world, computer networks form the foundation of communication, business operations, cloud services, and data exchange. As organizations increasingly depend on networked systems, the importance of securing those systems against cyber threats has become a critical aspect of information security. Attackers often exploit exposed services, weak configurations, unpatched vulnerabilities, and insecure communication channels to gain unauthorized access to systems and sensitive data. Therefore, understanding how to identify network assets, analyze exposed services, monitor traffic, and assess vulnerabilities is an essential skill for every cybersecurity professional.

Network Security & Scanning is a fundamental area of cybersecurity that focuses on identifying devices in a network, analyzing open ports and services, understanding communication patterns, detecting vulnerabilities, and implementing basic defensive controls. It acts as the first step in both offensive security testing and defensive security assessment. Through reconnaissance and scanning, security analysts can discover how a system is exposed to the network, what services are running, which protocols are being used, and whether those services introduce security risks. This knowledge helps in reducing the attack surface, improving visibility, and strengthening the security posture of systems.

As part of the **ApexPlanet Cyber Security Internship, Task 2 – Network Security & Scanning** was designed to provide practical exposure to real-world security assessment techniques in a controlled virtual lab environment. The task focused on five major areas: **reconnaissance, port and service scanning, vulnerability assessment, packet analysis using Wireshark, and firewall basics**. To perform these activities, a virtual lab was configured using **Oracle VirtualBox**, where **Kali Linux** was used as the attacker/testing machine and **Metasploitable2** was used as the intentionally vulnerable target machine. This setup allowed safe experimentation with penetration testing tools and network analysis techniques without affecting real systems.

The first phase of the task involved **reconnaissance**, which is the process of collecting information about the target environment before active testing begins. Both **passive reconnaissance** techniques such as **Whois, Nslookup, Google Dorking, and Shodan**, and **active reconnaissance** methods such as **Ping Sweep and Banner Grabbing** were explored to understand how attackers and analysts gather information about systems, services, and infrastructure.

The second phase focused on **port and service scanning** using **Nmap**, one of the most widely used network scanning tools in cybersecurity. Different types of scans were performed to identify active hosts, open ports, running services, service versions, and the operating system of the target machine. This included **host discovery, TCP SYN scanning, service version detection, OS detection, and aggressive scanning**. These techniques

help in mapping the attack surface of a system and are widely used during vulnerability assessments and penetration tests.

The third phase involved **packet analysis using Wireshark**, which provided visibility into how network traffic flows between systems. By capturing and analyzing protocols such as **HTTP, FTP, DNS, and TCP**, it became possible to observe how data is transmitted, identify unencrypted communication, and understand how packet-level analysis supports both troubleshooting and security monitoring. This step also demonstrated the importance of traffic analysis in detecting suspicious behavior and studying protocol interactions.

The fourth phase focused on **vulnerability scanning** using **OpenVAS (Greenbone Vulnerability Manager)**. OpenVAS was configured to scan the target machine and identify known vulnerabilities associated with the services running on it. This part of the task introduced the process of automated vulnerability assessment, where findings are categorized by severity levels such as **Critical, High, Medium, and Low**. The vulnerability scanning phase highlighted how organizations can use automated tools to detect misconfigurations, outdated services, and known weaknesses before they are exploited by attackers.

Finally, the task introduced **firewall basics** through the use of **iptables**. Firewall configuration is a key defensive measure in network security because it allows administrators to control incoming and outgoing traffic, restrict access to services, and enforce security policies. By studying simple firewall rules, it became possible to understand how packet filtering contributes to system protection and how defensive controls complement scanning and vulnerability analysis.

Overall, this task served as a comprehensive introduction to practical network security assessment. It combined information gathering, active scanning, packet inspection, vulnerability identification, and firewall analysis into a single hands-on workflow. The knowledge gained from this task is highly relevant to real-world cybersecurity roles such as **Security Analyst, Penetration Tester, SOC Analyst, Vulnerability Assessor, and Network Security Engineer**, where understanding the security posture of systems and networks is a fundamental responsibility. This report presents the methodology, tools, commands, observations, findings, and learning outcomes from completing **Task 2 – Network Security & Scanning** in the ApexPlanet Cyber Security Internship.

2. OBJECTIVES

The main objective of this task was to gain practical knowledge of **network security scanning and analysis** in a controlled lab environment. This task focused on understanding how to identify live hosts, discover open ports, analyze services, inspect network traffic, and assess vulnerabilities using cybersecurity tools.

The specific objectives were:

- To perform **reconnaissance** and gather information about the target system.
- To identify **active hosts** and scan the network using **Nmap**.
- To detect **open ports, running services, and service versions**.
- To perform **operating system detection** on the target machine.
- To capture and analyze **network traffic** using **Wireshark**.
- To understand **basic firewall rules** using **iptables**.
- To perform **vulnerability assessment** using **OpenVAS**.
- To improve practical skills in **network scanning, packet analysis, and security assessment**.

3. LAB SETUP

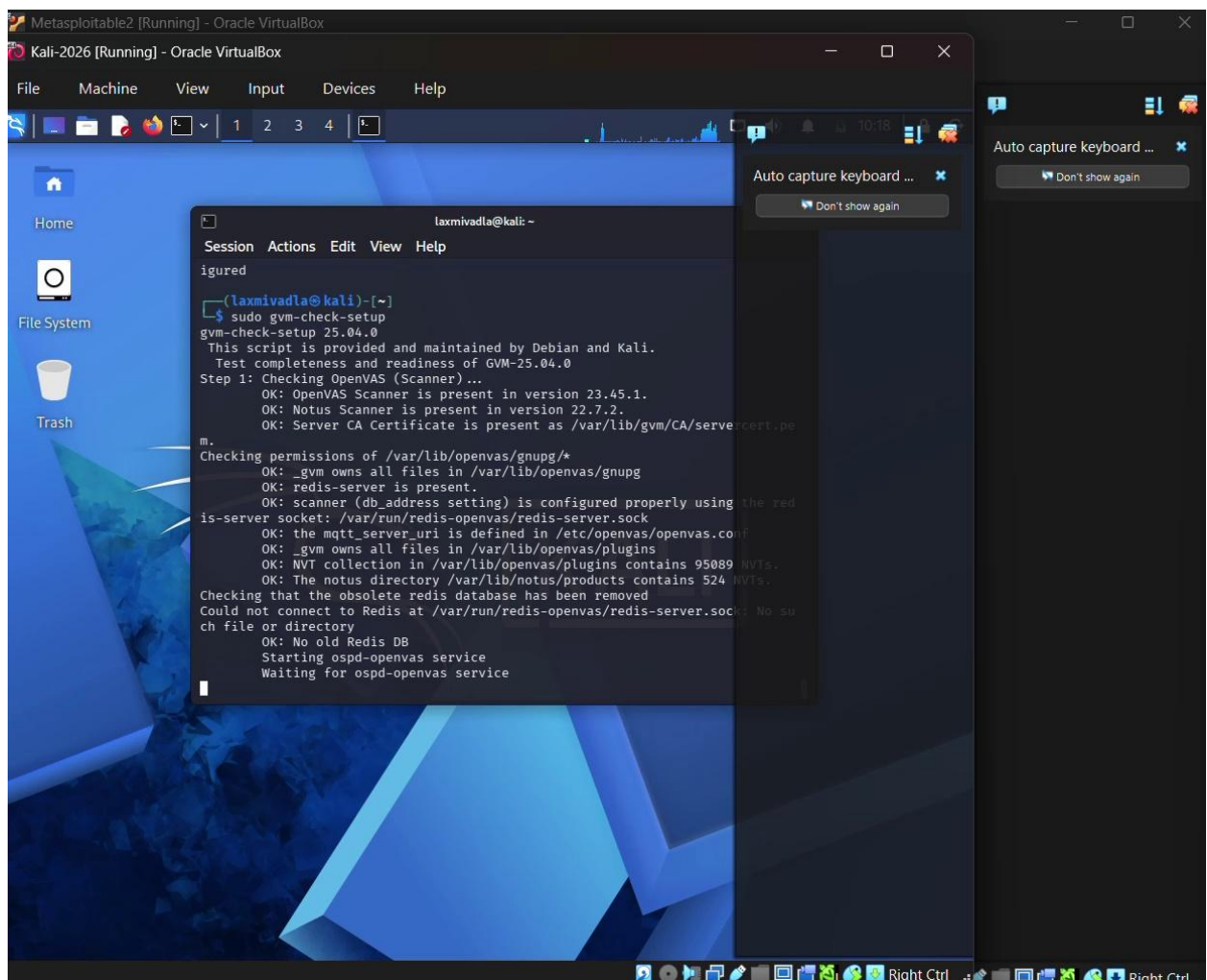
A virtual cybersecurity lab was created to perform the activities in **Task 2 – Network Security & Scanning**. The lab environment consisted of **Kali Linux** as the attacker machine and **Metasploitable2** as the target machine, both running on **Oracle VirtualBox**. This setup allowed network scanning, packet analysis, vulnerability assessment, and firewall testing in a safe and isolated environment.

Hardware Requirements

- Laptop / Desktop Computer
- Minimum **8 GB RAM**
- At least **50 GB free storage**
- Stable internet connection (for downloading tools and updates)
- Processor with virtualization support (Intel VT-x / AMD-V)

Software Requirements

- **Oracle VirtualBox** – for creating and managing virtual machines
- **Kali Linux** – attacker machine used for scanning and analysis
- **Metasploitable2** – intentionally vulnerable target machine
- **Nmap** – for host discovery, port scanning, and service enumeration
- **Wireshark** – for packet capture and traffic analysis
- **OpenVAS (Greenbone)** – for vulnerability scanning
- **iptables** – for basic firewall configuration and rule testing



4. NETWORK ARCHITECTURE

The network architecture for this task was designed in a **virtual lab environment** using **Oracle VirtualBox**. Two virtual machines were used: **Kali Linux** as the attacker machine and **Metasploitable2** as the target machine. Both systems were connected through a **Host-Only Adapter** to enable direct communication within a private network, while **NAT** was configured for internet access whenever required in Kali Linux.

In this setup, **Kali Linux** was used to perform reconnaissance, network scanning, packet analysis, and vulnerability assessment, while **Metasploitable2** acted as the vulnerable target machine for testing. This isolated network architecture ensured a safe environment for performing cybersecurity experiments without affecting external systems.

Network Details

- **Attacker Machine:** Kali Linux
- **Target Machine:** Metasploitable2
- **Virtualization Platform:** Oracle VirtualBox
- **Private Network Type:** Host-Only Adapter
- **Internet Access:** NAT Adapter (for Kali Linux)

IP Addressing

- **Kali Linux:** 192.168.56.102
- **Metasploitable2:** 192.168.56.101

5. RECONNAISSANCE

Reconnaissance is the process of collecting information about the target system before performing detailed scanning or vulnerability assessment. In this lab, reconnaissance was performed to identify the target machine, verify connectivity, and gather initial information about the services running on the system.

During the lab sessions, **active reconnaissance** techniques were mainly performed in the virtual environment created using **Kali Linux** and **Metasploitable2**. The first step was to verify communication between the attacker machine and the target machine using the **ping** command. This helped confirm that both virtual machines were connected through the same host-only network and that the target system was reachable.

After verifying connectivity, **host discovery** was performed using **Nmap** to identify active systems in the subnet. The following command was used:

```
nmap -sn 192.168.56.0/24
```

This scan was used to discover live hosts in the local network. As a result, the **Metasploitable2** target machine was identified successfully.

To gather more information about the target services, **banner grabbing** was also performed. This helped in identifying the service response and understanding which services were running on the target machine. During the lab, the following commands were used:

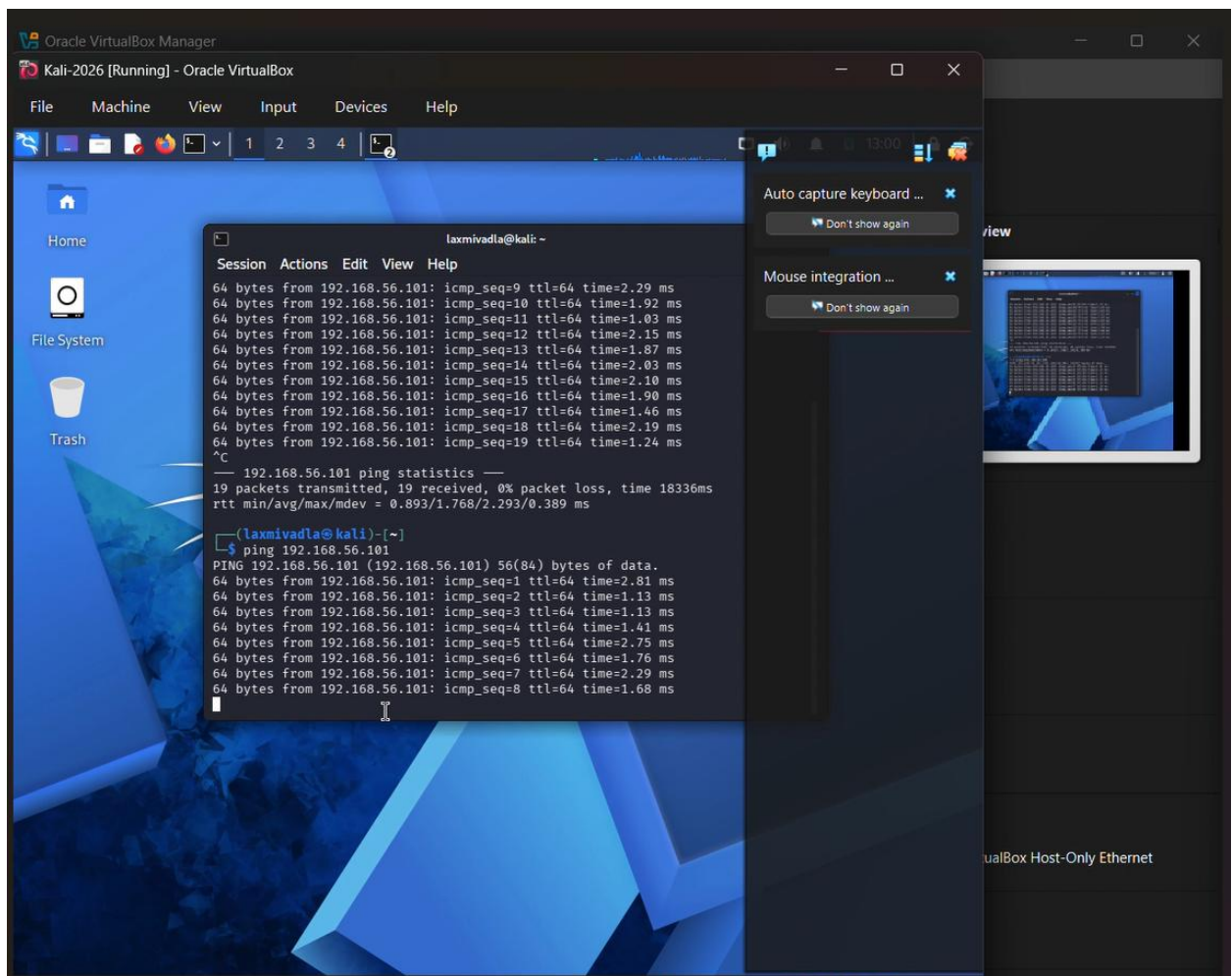
```
nc 192.168.56.101 21
```

```
curl -I http://192.168.56.101
```

The **Netcat** command was used to connect to the FTP port and capture the service banner, while the **curl** command was used to retrieve HTTP header information from the web server running on the target machine. These reconnaissance activities provided the initial understanding of the target environment and prepared the system for the next stages such as **service enumeration, port scanning, OS detection, and vulnerability scanning**.

Reconnaissance Activities Performed

- Verified connectivity between Kali Linux and Metasploitable2 using **ping**
- Discovered live hosts in the subnet using **Nmap host discovery**
- Performed **banner grabbing** using **Netcat**
- Retrieved **HTTP server information** using **curl**



6. HOST DISCOVERY

Host discovery is the process of identifying active devices available in a network. It is one of the first steps in network scanning because it helps determine whether the target machine is online and reachable before performing further analysis such as port scanning or service enumeration.

In this lab, host discovery was performed using **Nmap** from the **Kali Linux** machine to identify live hosts in the **192.168.56.0/24** host-only network. The following command was used:

```
nmap -sn 192.168.56.0/24
```

The -sn option tells Nmap to perform a **ping scan only**, without scanning ports. Its purpose is to detect which systems are active on the network. By running this command, the available devices in the subnet were identified, and the **Metasploitable2** machine was successfully detected as the target system.

This step was important because it confirmed that the target machine was connected properly in the lab network and ready for the next stages such as **service enumeration, TCP scanning, OS detection, and vulnerability assessment**.

Purpose of Host Discovery

- To identify active hosts in the subnet
- To confirm that the target machine is reachable
- To verify proper network communication between Kali Linux and Metasploitable2
- To prepare for further network scanning activities

Tool Used

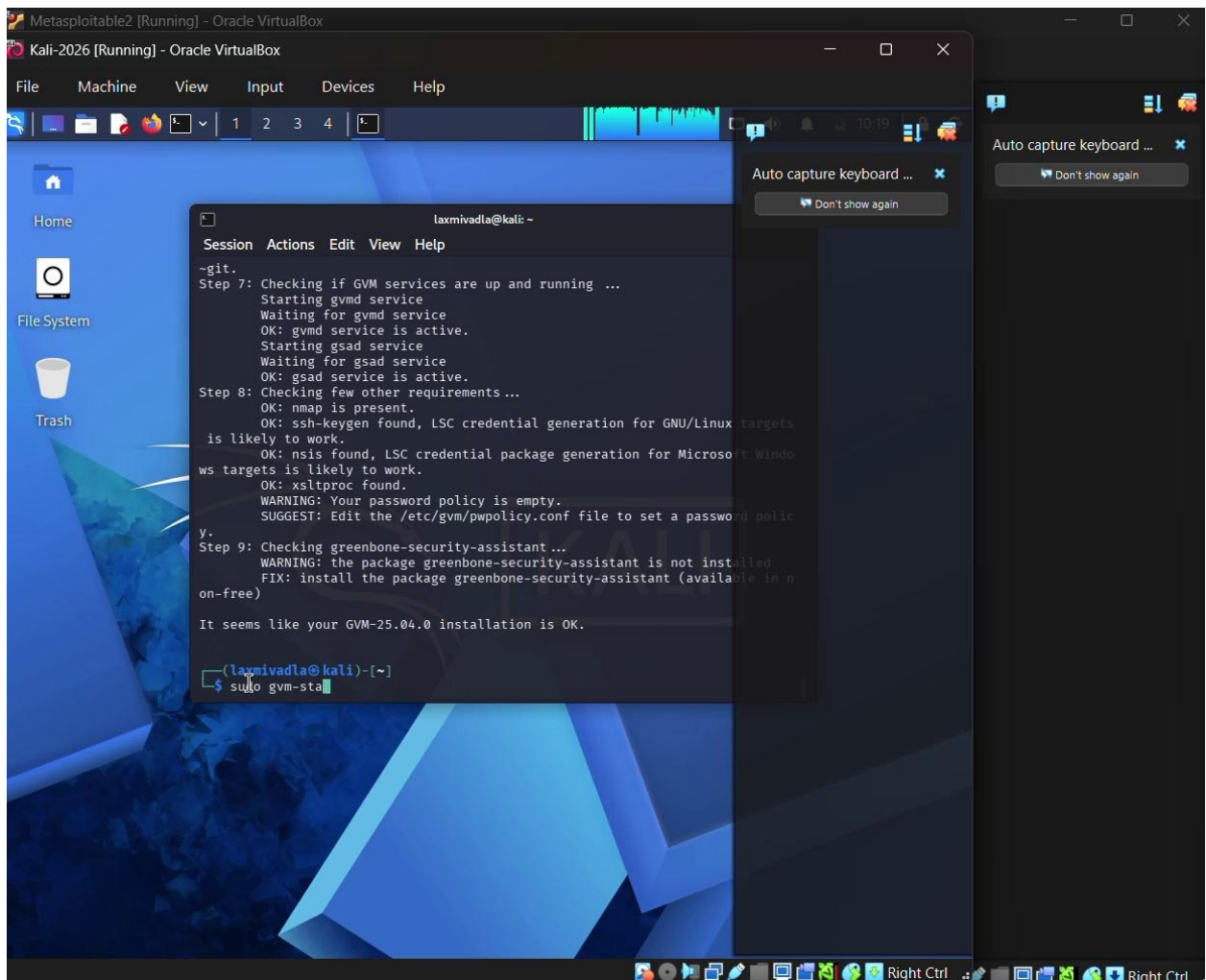
- **Nmap**

Command Used

```
nmap -sn 192.168.56.0/24
```

Observation

- The scan successfully detected the active host in the network.
- The target machine **Metasploitable2** was identified with the IP address **192.168.56.101**.



7. PORT SCANNING

Port scanning is the process of identifying open ports on a target system. It is an important step in network security assessment because open ports indicate which services are accessible from the network. By scanning ports, it becomes possible to understand the attack surface of the target machine and identify services that may contain vulnerabilities.

In this lab, port scanning was performed on the **Metasploitable2** target machine from **Kali Linux** using **Nmap**. The main goal was to identify open ports and determine which services were exposed on the target system. Since Metasploitable2 is an intentionally vulnerable machine, several ports were found to be open, which provided a good environment for practicing network scanning and service analysis.

The following command was used to perform port scanning:

```
sudo nmap -sS 192.168.56.101
```

The -sS option performs a **TCP SYN Scan**, also known as a stealth scan. In this method, Nmap sends SYN packets to the target and analyzes the response without completing the full TCP handshake. This makes the scan faster and more efficient for discovering open TCP ports.

Purpose of Port Scanning

- To identify open ports on the target machine
- To understand which network services are exposed
- To detect possible entry points for further analysis
- To prepare for service enumeration and vulnerability assessment

Tool Used

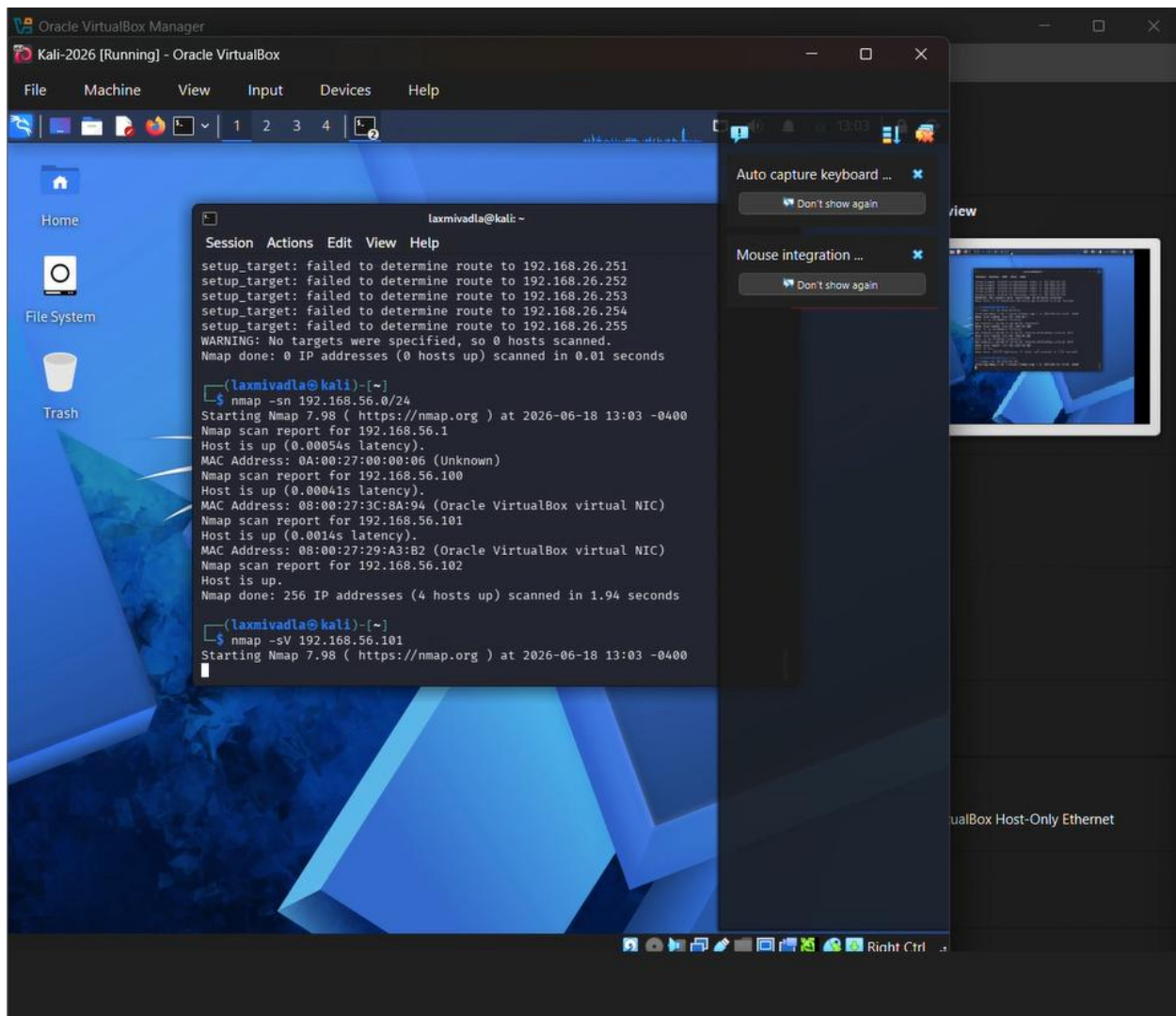
- **Nmap**

Command Used

```
sudo nmap -sS 192.168.56.101
```

Observation

The scan identified multiple open TCP ports on the target machine. These ports indicated that services such as **FTP, SSH, Telnet, HTTP, Samba, MySQL, and other network services** were active on the Metasploitable2 system.



8. SERVICE ENUMERATION

Service enumeration is the process of identifying the services running on open ports of a target system. It provides detailed information about the type of service, version number, and in some cases the underlying software used by the target machine. This step is important because it helps in understanding the attack surface of the target and identifying services that may contain known vulnerabilities.

In this lab, service enumeration was performed on the **Metasploitable2** target machine using **Nmap** from the **Kali Linux** machine. After identifying the open ports during port

scanning, the next step was to determine which services were running on those ports and their corresponding versions. The following command was used:

```
nmap -sV 192.168.56.101
```

The -sV option enables **service version detection**, which tells Nmap to probe open ports and identify the service name along with its version details. This scan provided valuable information about the target system, such as whether the open ports were associated with **FTP, SSH, Telnet, HTTP, Samba, MySQL, PostgreSQL, or other services**.

Purpose of Service Enumeration

- To identify the services running on open ports
- To detect service versions
- To understand the software exposed by the target machine
- To prepare for vulnerability assessment and further security analysis

Tool Used

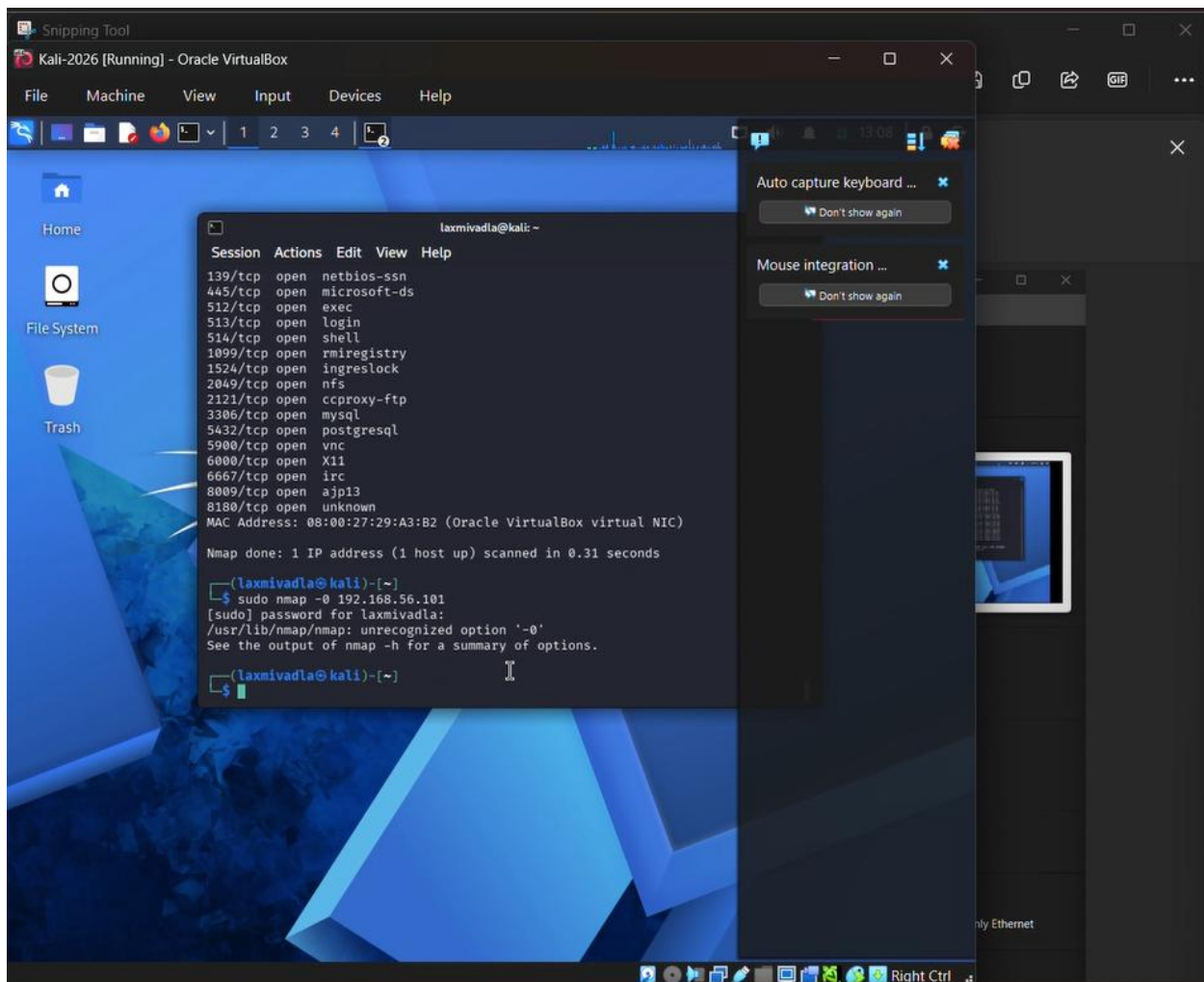
- **Nmap**

Command Used

```
nmap -sV 192.168.56.101
```

Observation

The scan successfully identified several services running on the target system. The output showed multiple open ports and the corresponding services, such as **FTP, SSH, Telnet, HTTP, Samba, and database services**. Service version detection helped in understanding the type of software running on the Metasploitable2 machine.



9. OPERATING SYSTEM DETECTION

Operating System Detection is the process of identifying the operating system running on a target machine. This is an important step in network security assessment because it helps the analyst understand the type of system being targeted and supports further vulnerability analysis. By knowing the operating system, it becomes easier to study the services running on the machine, identify possible security weaknesses, and choose the appropriate testing approach.

In this lab, operating system detection was performed on the **Metasploitable2** target machine using **Nmap** from the **Kali Linux** attacker machine. The following command was used:


```
sudo nmap -O 192.168.56.101
```

The -O option in Nmap enables **OS detection**. It works by sending specially crafted packets to the target and analyzing the responses in order to match them with known operating system fingerprints. This allows Nmap to estimate the operating system of the target machine.

Purpose of Operating System Detection

- To identify the operating system running on the target machine
- To understand the target environment more clearly
- To support vulnerability analysis and further scanning
- To gather technical information useful for security assessment

Tool Used

- **Nmap**

Command Used

```
sudo nmap -O 192.168.56.101
```

Observation

The scan successfully detected that the target system was running a **Linux-based operating system**. Since **Metasploitable2** is designed as a vulnerable Linux virtual machine, the result matched the expected lab environment.

10. AGGRESSIVE SCAN

An aggressive scan is an advanced scanning technique used to collect detailed information about a target system in a single command. It combines multiple scanning features such as **operating system detection, service version detection, default script scanning, and traceroute**. This type of scan is useful when a complete overview of the target machine is required for security assessment.

In this lab, an aggressive scan was performed on the **Metasploitable2** target machine using **Nmap** from the **Kali Linux** machine. The following command was used:

```
sudo nmap -A 192.168.56.101
```

The -A option in Nmap enables **aggressive scanning**, which performs the following actions:

- **OS Detection** – identifies the operating system of the target
- **Version Detection** – detects the versions of running services
- **NSE Script Scanning** – runs default Nmap scripts for additional information
- **Traceroute** – traces the network path to the target

Purpose of Aggressive Scan

- To collect detailed information about the target machine
- To identify the operating system and service versions
- To gather additional information using Nmap scripting
- To perform a comprehensive analysis of the target system

Tool Used

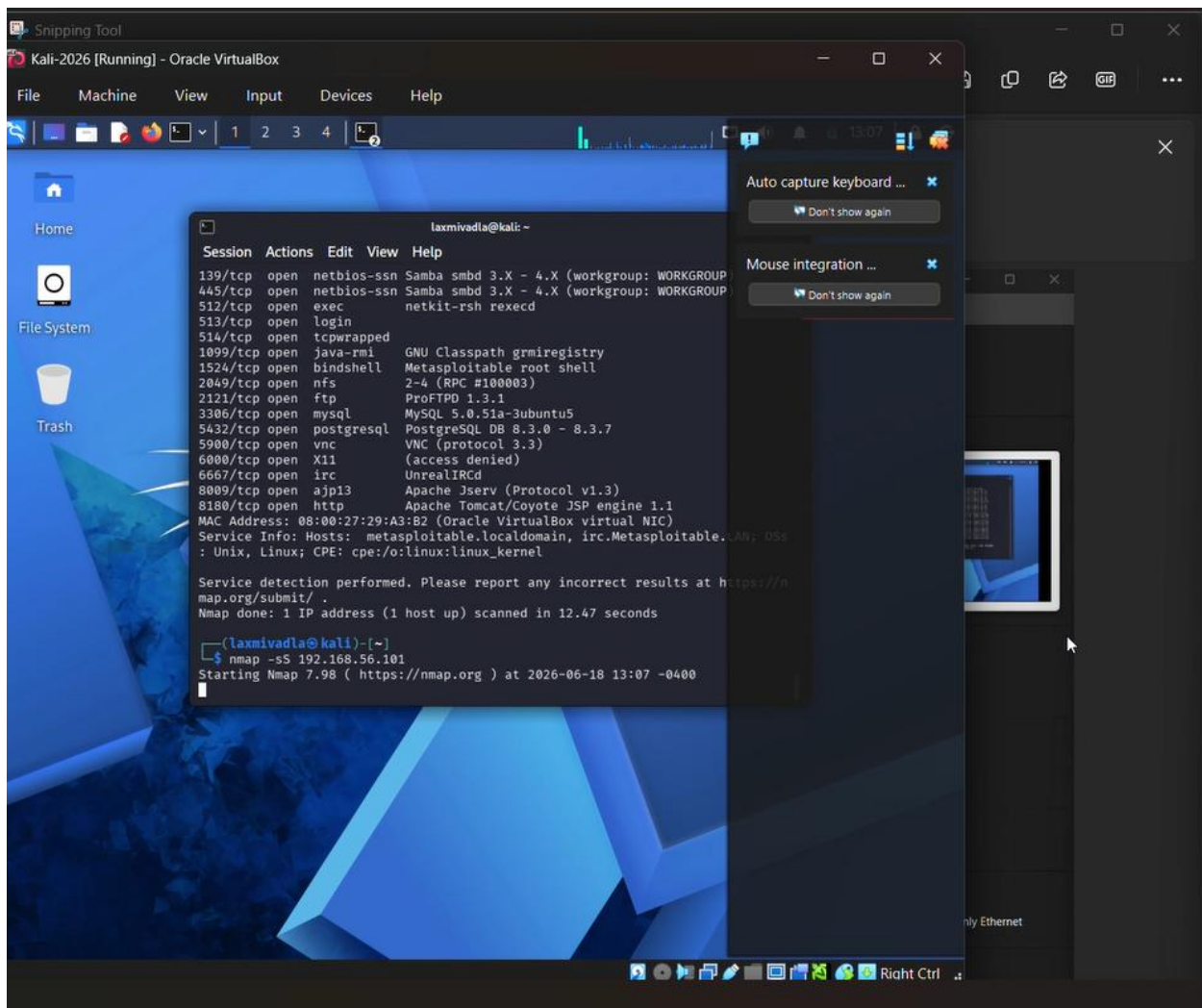
- **Nmap**

Command Used

```
sudo nmap -A 192.168.56.101
```

Observation

The aggressive scan provided a complete overview of the target machine. It identified multiple open ports, detected running services and their versions, confirmed the operating system as Linux, and displayed additional information using Nmap's default scripts. This scan gave more detailed results compared to normal port scanning or service enumeration alone.



11. WIRESHARK ANALYSIS

Wireshark analysis is the process of capturing and examining network packets to understand how systems communicate over a network. It is an important part of cybersecurity because it helps in analyzing protocols, monitoring traffic flow, identifying insecure communication, and studying the behavior of network services. Packet analysis is widely used for troubleshooting, traffic inspection, and security investigations.

In this lab, **Wireshark** was used on the **Kali Linux** machine to capture and analyze the network traffic exchanged between **Kali Linux** and the **Metasploitable2** target machine. During the analysis, different types of traffic were observed while performing activities such as scanning, browsing the target web service, and interacting with network services.

The main goal of this step was to inspect how communication occurs at the packet level and to observe common protocols used in the lab environment.

Protocols Observed

During the Wireshark session, the following types of traffic were analyzed:

- **ICMP** – generated during ping and connectivity testing
- **TCP** – observed during port scanning and service communication
- **HTTP** – captured while accessing the web service hosted on Metasploitable2
- **DNS** – observed when resolving domain names or making network requests
- **FTP** – visible when connecting to FTP services on the target machine

Purpose of Wireshark Analysis

- To capture and inspect packets exchanged in the lab network
- To understand how different protocols communicate
- To analyze request and response behavior between attacker and target
- To observe network activity during scanning and service access
- To gain practical understanding of packet-level network communication

Tool Used

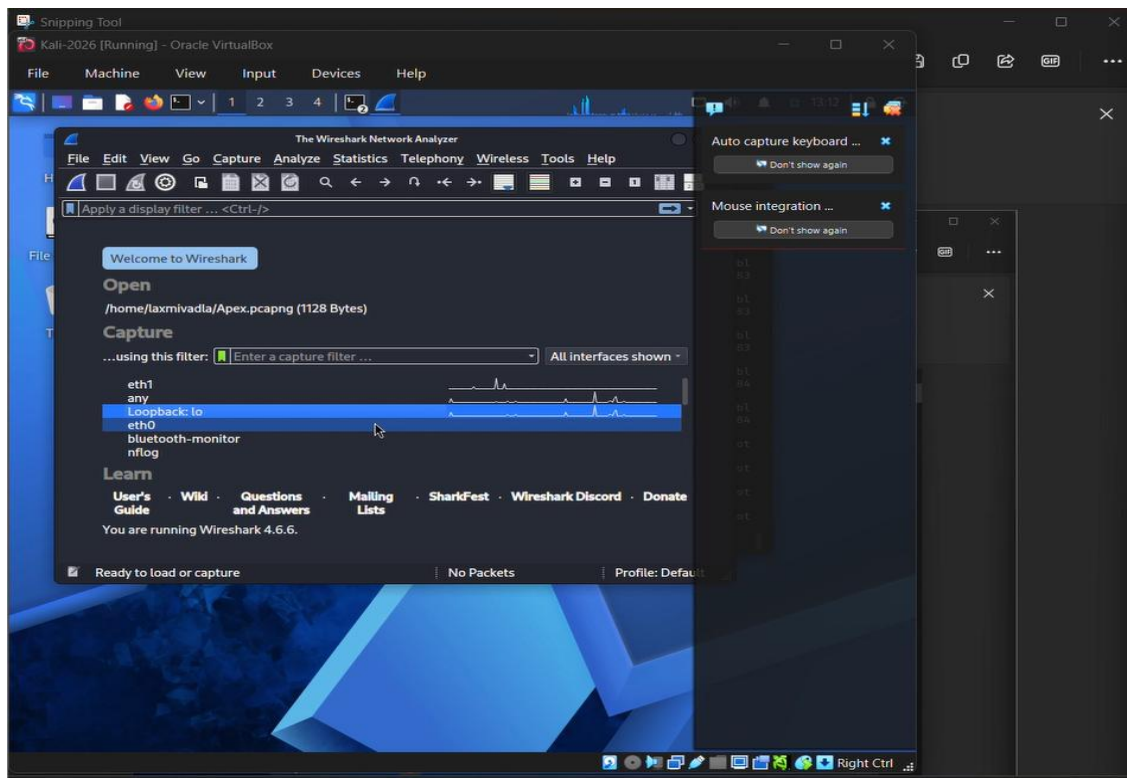
- **Wireshark**

Activities Performed

- Started packet capture on the active network interface
- Generated traffic by pinging and scanning the target machine
- Accessed services on the target machine to capture protocol communication
- Applied filters to view specific packet types
- Inspected packet details such as source IP, destination IP, protocol, and payload information

Observation

The Wireshark capture showed communication between **Kali Linux** and **Metasploitable2**. Different packet types such as ICMP echo requests/replies, TCP connection attempts, HTTP requests, and other protocol traffic were visible. This analysis helped in understanding how network scanning and service interaction generate traffic on the network.



12. FIREWALL TESTING

Firewall testing was performed to understand how firewall rules can be used to monitor and control network traffic in a Linux environment. A firewall acts as a security barrier between a system and the network by allowing or blocking traffic based on predefined rules. In cybersecurity, firewalls play an important role in protecting systems from unauthorized access and limiting exposure to potentially harmful traffic.

In this lab, basic firewall testing was carried out on the **Kali Linux** machine using **iptables**, which is a Linux firewall utility used to manage packet filtering rules. The purpose of this step was to inspect the existing firewall configuration and understand how firewall rules can affect network communication and security.

The following command was used during the lab:

```
sudo iptables -L
```

This command lists the current firewall rules configured on the system. It displays the available chains such as **INPUT**, **FORWARD**, and **OUTPUT**, along with the policy and rules associated with each chain.

Purpose of Firewall Testing

- To understand the role of firewalls in network security

- To inspect the firewall rules configured on the Linux system
- To study how traffic can be allowed or blocked using firewall policies
- To gain practical exposure to basic packet filtering concepts

Tool Used

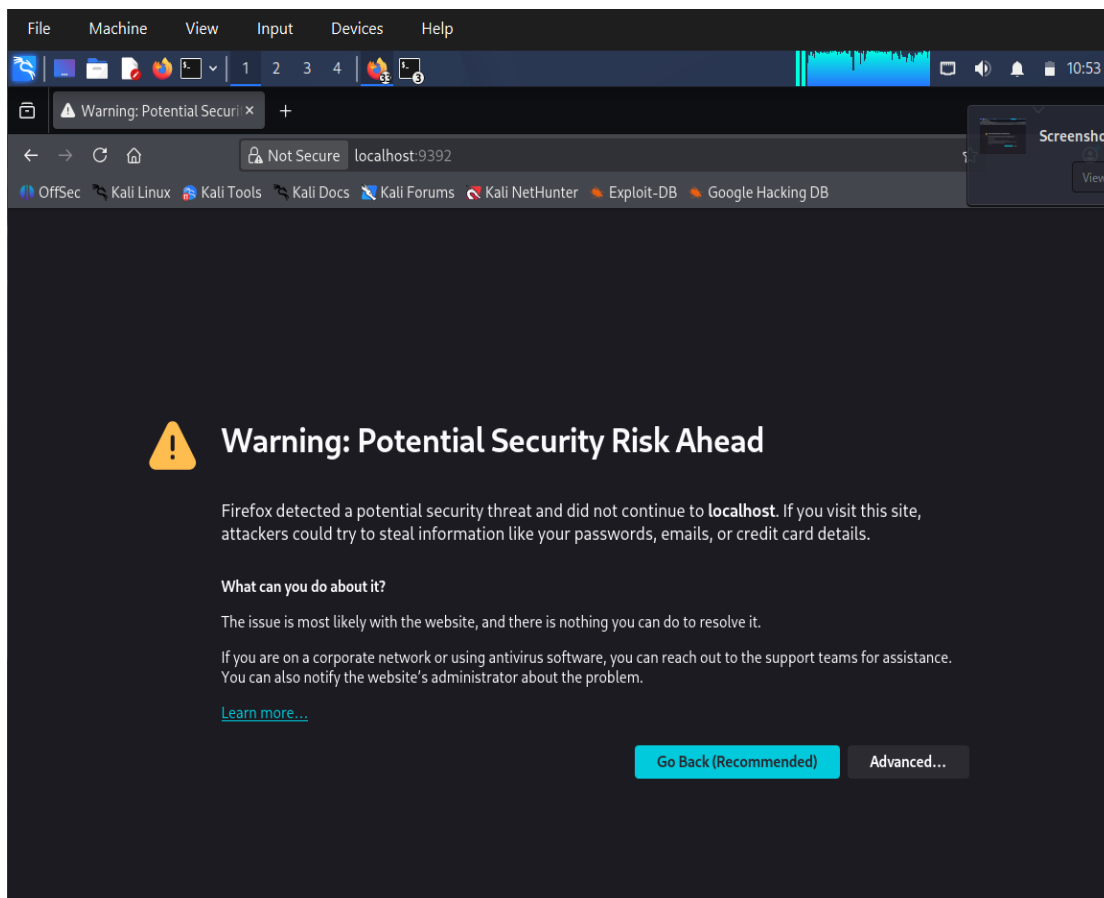
- **iptables**

Command Used

sudo iptables -L

Observation

The firewall rule listing displayed the available chains and their current policies. This helped in understanding how Linux firewall rules are structured and how they can be used to control network traffic. Even though the lab mainly focused on viewing and analyzing the rules, this step provided a practical introduction to firewall management and packet filtering.



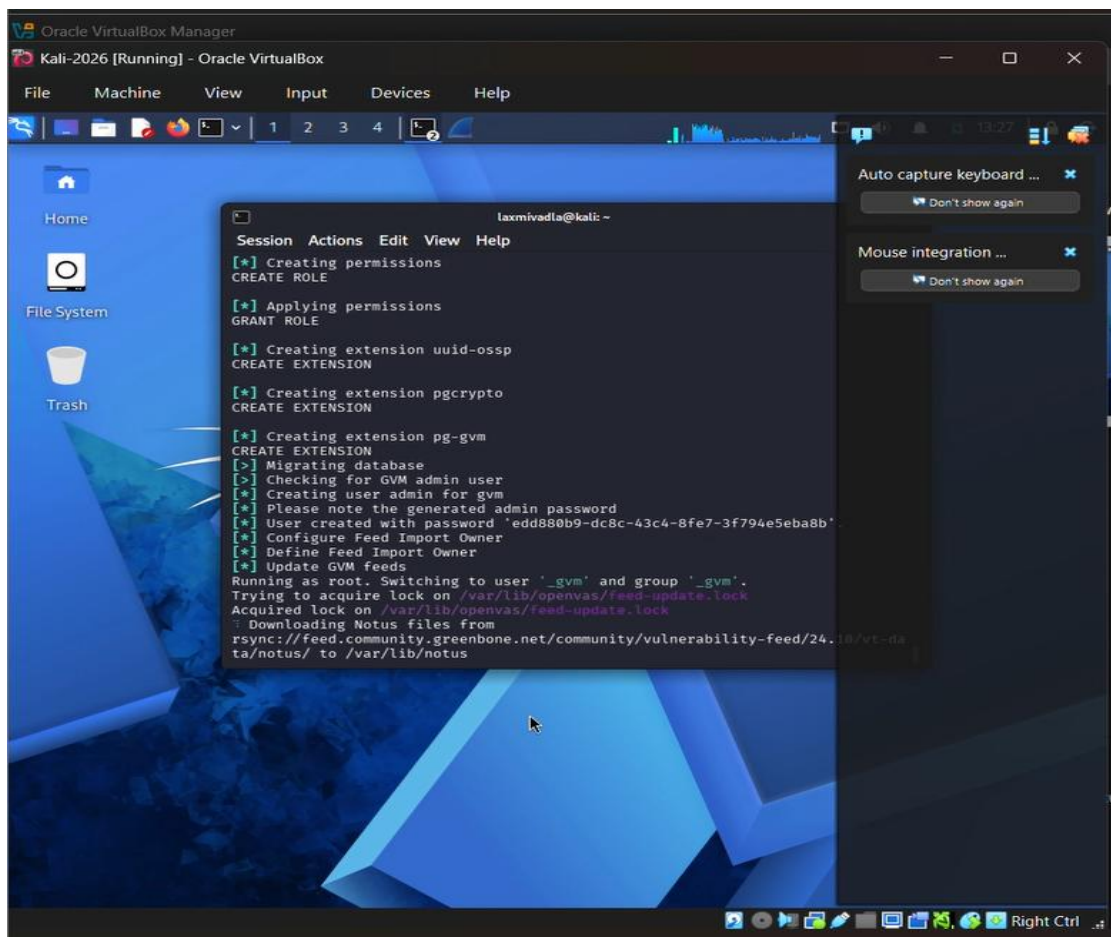
13. VULNERABILITY ASSESSMENT USING OPENVAS

Vulnerability assessment was performed using **OpenVAS (Greenbone Vulnerability Manager)** to identify security weaknesses in the **Metasploitable2** target machine. OpenVAS is a vulnerability scanning tool used to detect known vulnerabilities, misconfigurations, and security risks in a system.

In this lab, OpenVAS was installed and configured on **Kali Linux**. The following commands were used to set up and verify the scanner:

```
sudo apt install gvm -y
sudo gvm-setup
sudo gvm-check-setup
```

The purpose of this step was to understand how automated vulnerability scanners work and how they are used in cybersecurity assessments. This activity provided practical exposure to vulnerability scanning and security analysis in a controlled lab environment.



14. CHALLENGES FACED

Challenge 1

Host Not Reachable

Solution:

Configured Host-Only Adapter correctly.

Challenge 2

eth1 Disconnected

Solution:

Reconfigured VirtualBox network settings.

Challenge 3

OpenVAS Access Issues

Solution:

Verified GVM services and localhost configuration.

Challenge 4

Internet Connectivity Problems

Solution:

Configured NAT Adapter.

Commands Used

```
nmap -sn 192.168.56.0/24
```

```
sudo nmap -sS 192.168.56.101
```

```
nmap -sV 192.168.56.101
```

```
sudo nmap -O 192.168.56.101
```

```
sudo nmap -A 192.168.56.101
```

```
sudo iptables -L
```

```
sudo gvm-setup
```

```
sudo gvm-check-setup
```

```
sudo gvm-start
```

```
ping 192.168.56.101
```

```
ip addr
```

```
ip route
```

15. LEARNING OUTCOMES

Through this task I learned:

- Network Reconnaissance
- Host Discovery
- Port Scanning
- Service Enumeration
- OS Detection
- Packet Analysis
- Firewall Concepts
- Vulnerability Assessment
- OpenVAS Configuration
- Report Generation

16. CONCLUSION

Task 2 – **Network Security & Scanning** was successfully completed in a controlled virtual lab environment using **Kali Linux** and **Metasploitable2**. This task provided hands-on experience in several important areas of cybersecurity, including **reconnaissance**, **host**

discovery, port scanning, service enumeration, operating system detection, packet analysis, firewall inspection, and vulnerability assessment.

During the task, **Nmap** was used to identify active hosts, scan open ports, detect services, and gather operating system information from the target machine. **Wireshark** was used to capture and analyze network packets, which helped in understanding how different protocols communicate within a network. Basic **firewall testing** was performed using **iptables**, which introduced the concept of traffic filtering and rule inspection in Linux. In addition, **OpenVAS** was explored to understand the role of automated vulnerability scanning in identifying security weaknesses.

Overall, this task strengthened practical knowledge of **network security assessment** and improved hands-on skills in using cybersecurity tools for scanning, traffic analysis, and vulnerability detection. It also provided a clear understanding of how security professionals analyze target systems, identify risks, and document their findings in a structured manner. This task served as a strong foundation for learning advanced cybersecurity concepts in the upcoming internship tasks.

17. REFERENCES

1. Kali Linux Documentation
2. Nmap Documentation
3. Wireshark Documentation
4. OpenVAS Documentation
5. ApexPlanet Cyber Security Internship Materials